

# Development Threat in the New Jersey Pinelands: Headwater Conflict and Noise Pollution

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## An Analysis of Ecological Risk from Large-Scale Development on State-Designated Pinelands Commission Lands

*Research, analysis, and figures by terpinedream*

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### Why the Pinelands Matter

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The New Jersey Pinelands is a vast region spanning seven counties in southern New Jersey. It contains one of the largest and most intact coastal forest ecosystems in the northeastern United States. Its sandy, acidic soils support globally rare plant and animal communities, including species found nowhere else on Earth. The region sits atop one of the most important aquifers on the East Coast, and its headwater streams feed rivers, bays, and estuaries across the region.

The Pinelands Commission, a state regulatory body established in 1979, oversees development within the state-designated Pinelands Area. This analysis covers only the state-designated Pinelands, approximately 940,000 acres, which falls under the jurisdiction of the Pinelands Commission. It does not include the broader Pinelands National Reserve, which extends into coastal areas east of the Garden State Parkway and around Atlantic City. Those areas are administered under separate NJDEP coastal protections.

The Commission divides the state-designated landscape into management areas that range from strict Preservation Area Districts, where almost no development is allowed, to Regional Growth Areas, where residential and commercial development is actively encouraged.

This report is organized in two parts. **Part I** identifies where development-zoned land overlaps with ecologically critical headwater areas and state-designated biodiversity sites, and evaluates which conflict parcels remain genuinely at risk after accounting for existing protections. **Part II** extends the analysis to examine how the acoustic footprint of large industrial facilities — particularly data centers — would degrade wildlife habitat far beyond any parcel boundary.

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## Part I: Headwater and Habitat Conflict

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## What We Analyzed

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We examined all parcels within the state-designated Pinelands Area, covering portions of seven New Jersey counties: Atlantic, Burlington, Camden, Cape May, Cumberland, Gloucester, and Ocean.

For every parcel, we determined:

1. **Land use classification** from NJ MOD-IV tax records (Vacant, Commercial/Industrial, Residential, Agriculture, Protected/Exempt)
  2. **Headwater zone status**, whether the parcel sits within 150 meters of a headwater stream (streams with drainage area of 1 km<sup>2</sup> or less)
  3. **Heritage site status**, whether the parcel overlaps a NJ Natural Heritage Priority biodiversity site
  4. **Management area**, the Pinelands Commission zoning designation that controls what can be built
  5. **Wetland buffer status**, whether the parcel falls within a wetland or the Commission's mandatory 300-foot buffer around wetlands
  6. **Permanent land protection status**, whether development rights have been sold or the parcel is deed-restricted
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## The State-Designated Pinelands at a Glance

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|                                        |                        |
|----------------------------------------|------------------------|
| Total parcels analyzed                 | 252,414                |
| Total area                             | ~940,000 acres         |
| Development-zoned parcels              | 32,105 (108,309 acres) |
| Residential parcels                    | 96,940 (108,259 acres) |
| Protected/Exempt parcels               | 28,308 (419,335 acres) |
| Agricultural parcels                   | 2,828 (75,817 acres)   |
| Parcels in headwater zones             | 49,190                 |
| Parcels in heritage biodiversity sites | 21,731                 |

Nearly half of the state-designated Pinelands (419,335 acres) is already in protected or exempt status, reflecting decades of conservation investment. However, 108,309 acres remain classified as development-zoned land, including both vacant parcels and existing commercial or industrial properties.

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## Management Areas: How the Commission Controls Development

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The Pinelands Commission assigns every parcel to a management area that determines what development is permitted. These range from the most protective to the most permissive:

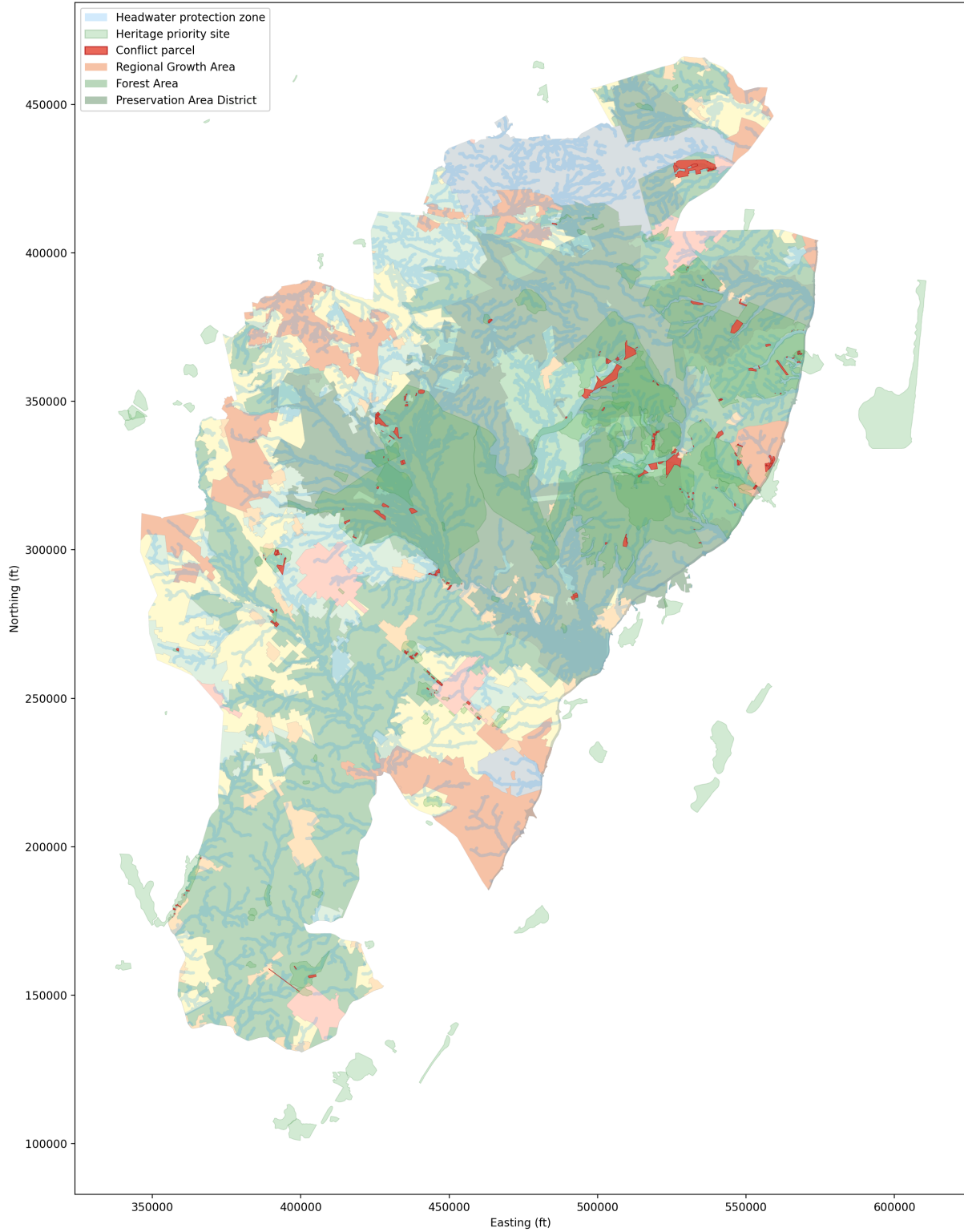
| Management Area               | Parcels | Acres   | Dev. Parcels | In Wetland Buffer | In PLP |
|-------------------------------|---------|---------|--------------|-------------------|--------|
| Preservation Area District    | 43,502  | 280,967 | 3,939        | 12,188            | 14,133 |
| Forest Area                   | 43,940  | 254,474 | 6,767        | 19,884            | 13,706 |
| Rural Development Area        | 37,711  | 98,811  | 5,640        | 17,854            | 5,195  |
| Regional Growth Area          | 84,509  | 69,450  | 9,682        | 27,686            | 1,056  |
| Agricultural Production Area  | 5,719   | 67,373  | 821          | 3,176             | 793    |
| Military/Federal Installation | 81      | 47,115  | 8            | 43                | 3      |
| Special AG Production Area    | 2,203   | 37,362  | 86           | 836               | 1,864  |
| Pinelands Village             | 17,875  | 22,661  | 2,842        | 4,862             | 530    |
| Pinelands Town                | 16,874  | 19,620  | 2,320        | 5,069             | 366    |

The Regional Growth Area has the most development-zoned parcels (9,682) within the state-designated boundary, reflecting its suburban landscape. The Preservation Area District and Forest Area together account for over 535,000 acres and form the ecological core of the Pinelands.

## Where Development Threatens Headwaters and Habitat

We identified **331 development-zoned parcels totaling 5,200 acres** that sit within both a headwater protection zone and a heritage biodiversity site. These are the conflict parcels — places where new development would simultaneously damage water resources and threaten rare species habitat.

# Pinelands Commission: Headwater & Heritage Conflict Analysis

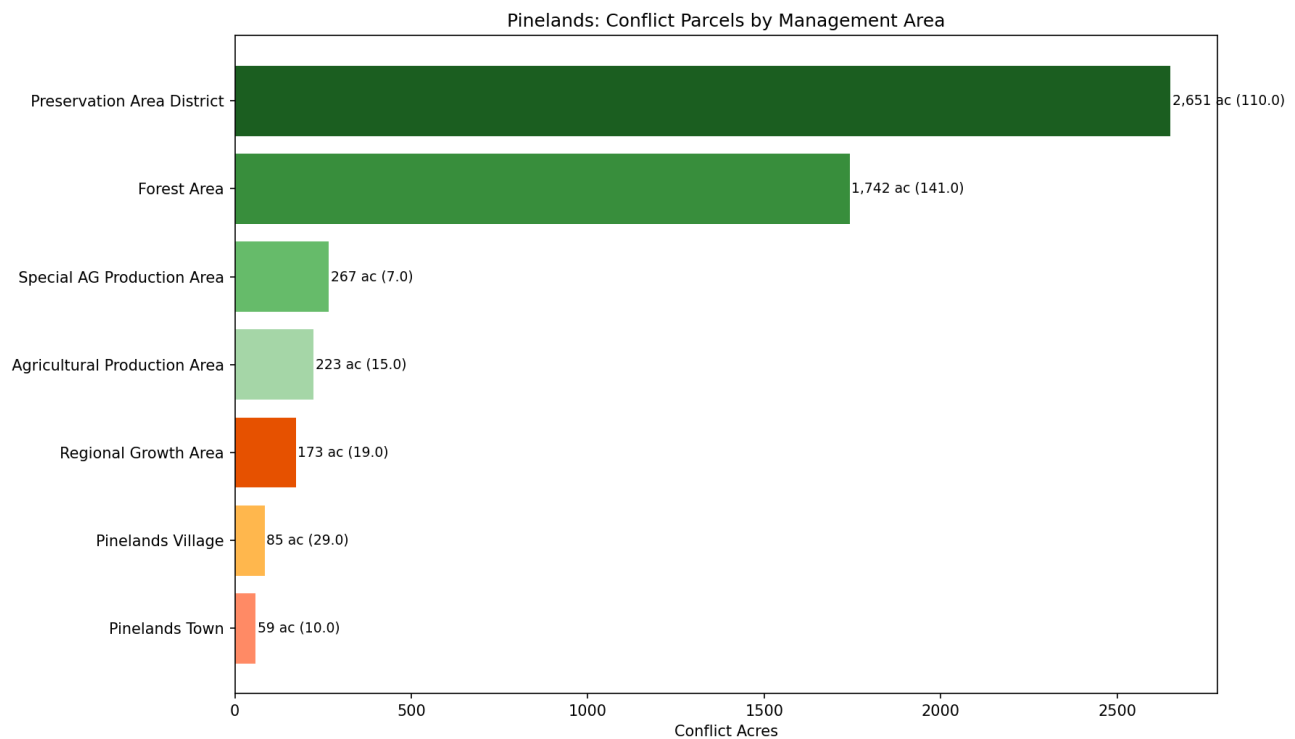


*Pinelands Commission: headwater and heritage conflict analysis with management area underlay*

## Conflict by management area

Not all conflicts carry the same urgency. A conflict parcel in a Regional Growth Area, where the Commission encourages development, is far more concerning than one in a Preservation Area, where development is already severely restricted.

| Management Area              | Conflict Parcels | Conflict Acres |
|------------------------------|------------------|----------------|
| Preservation Area District   | 110              | 2,651          |
| Forest Area                  | 141              | 1,742          |
| Special AG Production Area   | 7                | 267            |
| Agricultural Production Area | 15               | 223            |
| Regional Growth Area         | 19               | 173            |
| Pinelands Village            | 29               | 85             |
| Pinelands Town               | 10               | 59             |



*Pinelands: conflict parcels by management area*

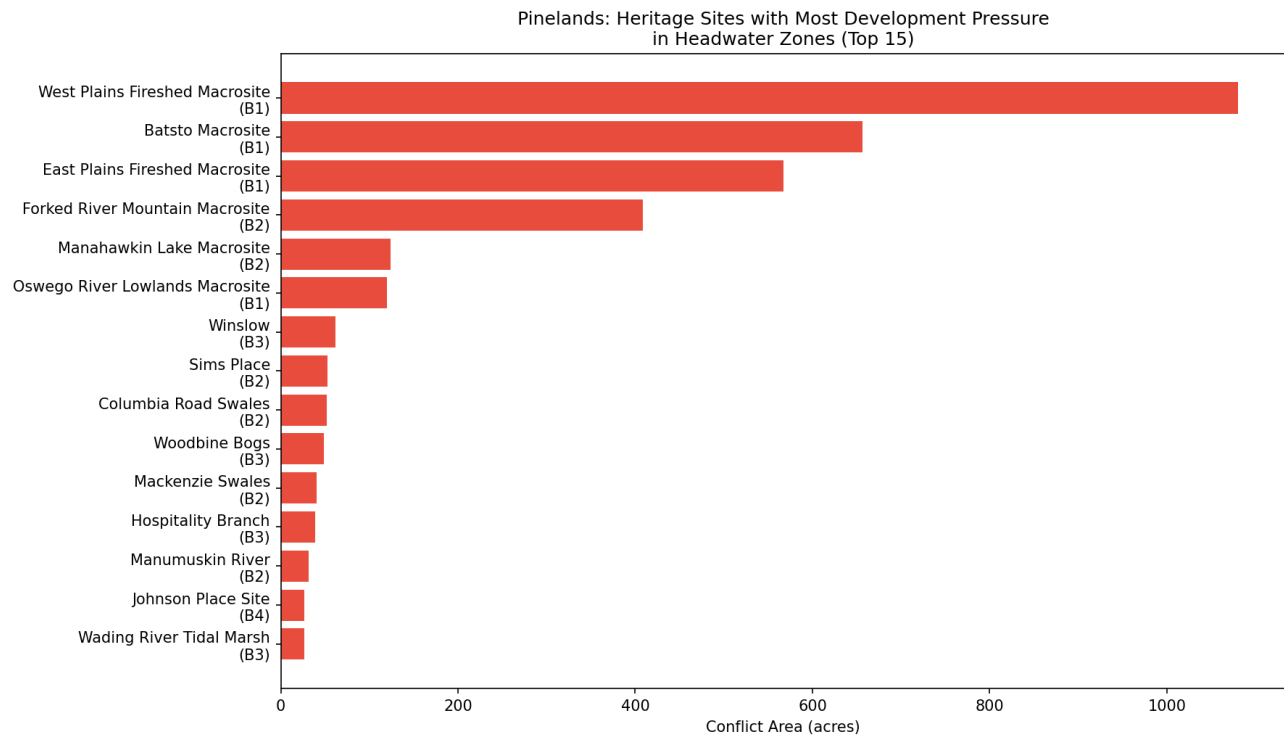
The largest conflict areas by acreage are in the Preservation Area District (2,651 acres) and Forest Area (1,742 acres), but these are also the zones where the Commission's regulations most restrict development. The 19 conflict parcels (173 acres) in the Regional Growth Area deserve the most attention because development there faces the fewest regulatory barriers.

## The Most At-Risk Heritage Sites

Across the state-designated Pinelands, 28 heritage biodiversity sites contain development-zoned parcels within headwater zones. The top conflict sites, ranked by acreage:

| Site                                               | Biodiversity Rank | Conflict Acres | Parcels | % Vacant | Assessed Value |
|----------------------------------------------------|-------------------|----------------|---------|----------|----------------|
| <b>West Plains<br/>Fireshed<br/>Macrosite</b>      | <b>B1</b>         | 1,080          | 54      | 98%      | \$1.2M         |
| <b>Batsto<br/>Macrosite</b>                        | <b>B1</b>         | 656            | 45      | 84%      | \$4.6M         |
| <b>East Plains<br/>Fireshed<br/>Macrosite</b>      | <b>B1</b>         | 568            | 23      | 96%      | \$2.0M         |
| <b>Forked<br/>River<br/>Mountain<br/>Macrosite</b> | B2                | 409            | 85      | 100%     | \$625K         |
| <b>Manahawkin<br/>Lake<br/>Macrosite</b>           | B2                | 124            | 11      | 91%      | \$726K         |
| <b>Oswego<br/>River<br/>Lowlands<br/>Macrosite</b> | <b>B1</b>         | 119            | 11      | 100%     | \$98K          |
| Winslow                                            | B3                | 61             | 13      | 100%     | \$113K         |
| Sims Place                                         | B2                | 53             | 1       | 100%     | \$101K         |
| Columbia<br>Road Swales                            | B2                | 52             | 9       | 89%      | \$232K         |
| Woodbine<br>Bogs                                   | B3                | 49             | 2       | 100%     | \$92K          |

Four of the top six conflict sites hold the state's highest biodiversity rank (B1), meaning they contain irreplaceable ecological resources. The West Plains Fireshed Macrosite alone has 1,080 acres of conflict land, 98% of which is still vacant.



*Pinelands: heritage sites ranked by development pressure in headwater zones*

## What Is Actually Developable? Applying Pinelands Protections

Raw conflict numbers overstate the true development threat because the Pinelands Commission enforces two additional protections that make many parcels effectively undevelopable:

1. **Wetlands + 300-foot buffer:** The Commission prohibits disturbance within wetlands and a 300-foot buffer around them. Any parcel within this zone cannot be built on.
2. **Permanent Land Protection Restrictions (PLP):** Parcels where development rights have been sold, or which carry deed restrictions, cannot be developed regardless of zoning.

After removing conflict parcels that fall within wetland buffers or PLP zones, we are left with the **net conflict**: parcels that are truly at risk.

|           | All Conflict | Net Conflict      |
|-----------|--------------|-------------------|
| Parcels   | 331          | 56                |
| Acres     | 5,200        | 855               |
| Reduction |              | 83% fewer parcels |

The Pinelands' layered protections eliminate **83% of conflict parcels** from genuine development risk. The 56 remaining net conflict parcels (855 acres) are the ones that matter most.

## Net conflict by management area

| Management Area            | Net Conflict Parcels | Net Conflict Acres | Vacant | Land Value | Assessed Value |
|----------------------------|----------------------|--------------------|--------|------------|----------------|
| Preservation Area District | 43                   | 696                | 42     | \$2.6M     | \$2.8M         |
| Forest Area                | 13                   | 159                | 12     | \$3.8M     | \$6.5M         |

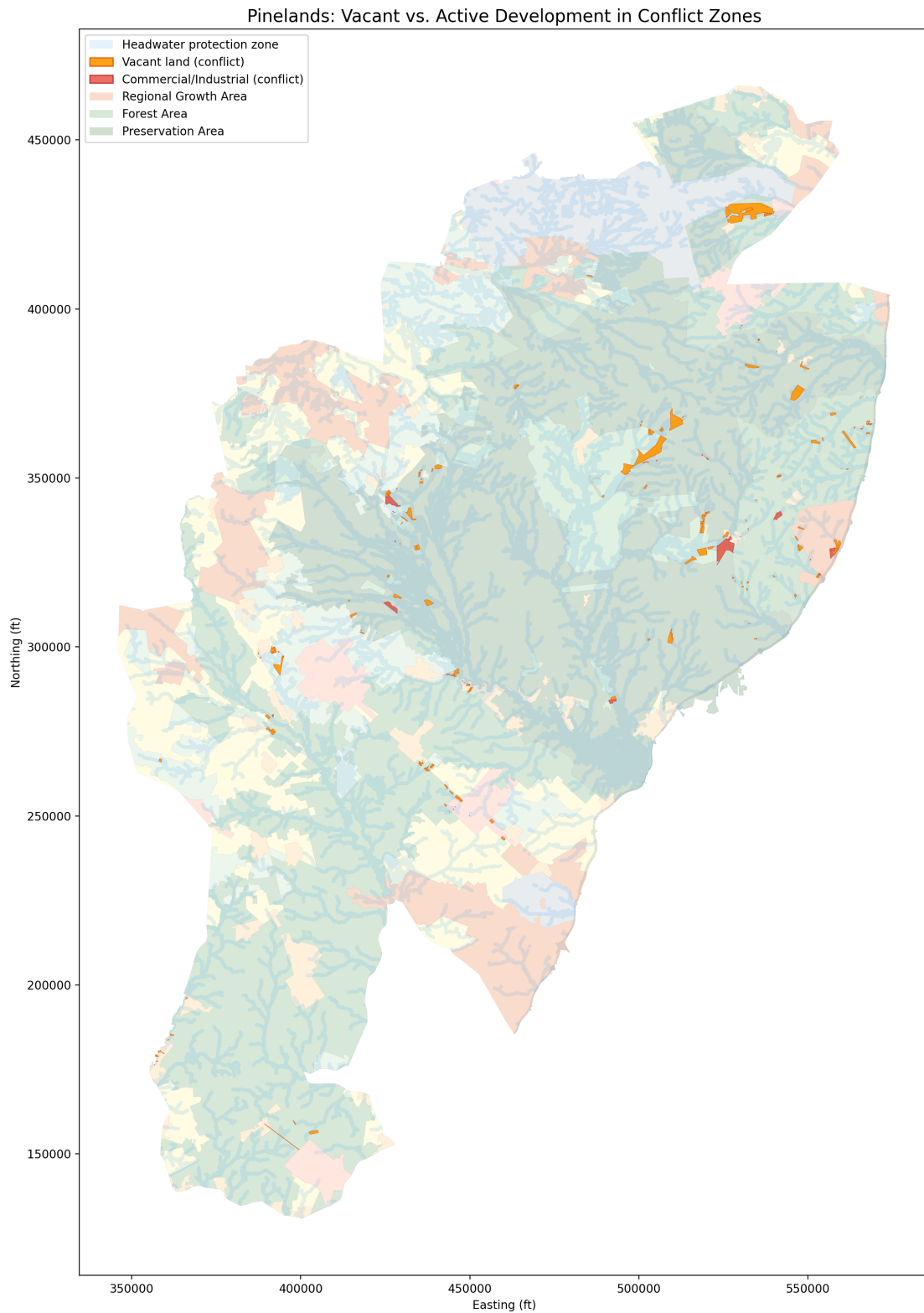
Notably, all 56 net conflict parcels fall within Preservation Area District or Forest Area, both heavily regulated management areas. The Regional Growth Area conflict parcels (19 total) were all eliminated by wetland buffers or PLP restrictions, meaning that the Commission's existing protections fully cover the most development-permissive zone. This is a significant finding: within the state-designated Pinelands, the regulatory framework is working effectively in the growth corridors.

## Vacant Land: The Conservation Window

Of the 331 conflict parcels, **316 (95%) are vacant**, meaning they are zoned for development but nothing has been built yet. This is the conservation window.

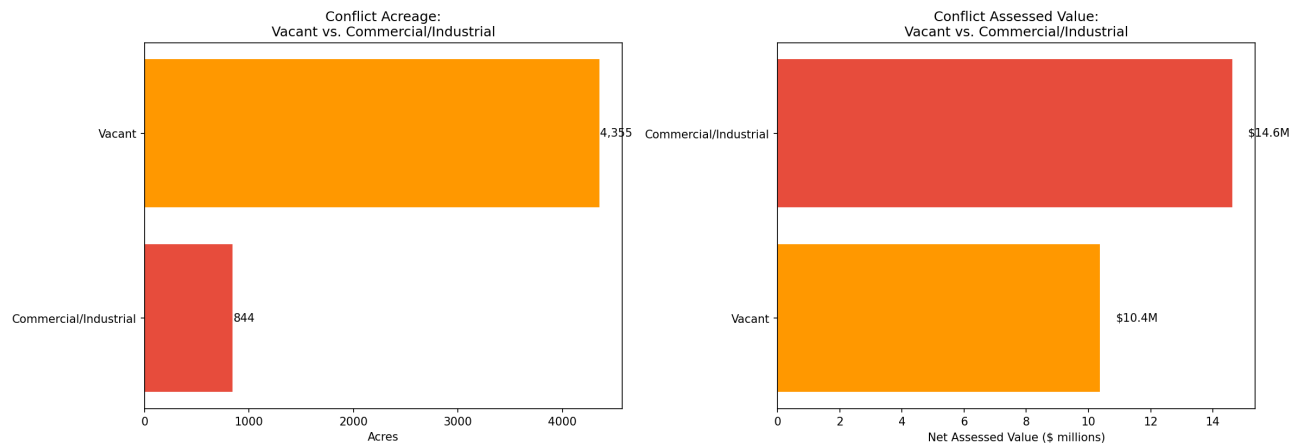
|                       | Parcels    | Acres        | Land Value     | Assessed Value |
|-----------------------|------------|--------------|----------------|----------------|
| Vacant                | 316        | 4,355        | \$10.4M        | \$10.4M        |
| Commercial/Industrial | 15         | 844          | \$9.2M         | \$14.6M        |
| <b>Total</b>          | <b>331</b> | <b>5,200</b> | <b>\$19.6M</b> | <b>\$25.0M</b> |

The 316 vacant conflict parcels cover 4,355 acres and carry a combined land value of **\$10.4 million**. That works out to approximately **\$2,400 per acre**, remarkably affordable for conservation acquisition. The low values reflect the wetland and sandy soil constraints that limit development potential in the Pinelands, even where zoning technically allows it.



*Pinelands: vacant vs. active development in conflict zones*

#### Pinelands Commission: Development Conflict Breakdown



Pinelands: development conflict breakdown by acreage and assessed value

## Highest-Priority B1 Sites

Sites with the state's highest biodiversity rank (B1) contain the most irreplaceable ecological resources. Four B1 sites in the state-designated Pinelands have significant development conflict in headwater zones:

| B1 Site                         | Conflict Acres | % Vacant | Assessed Value |
|---------------------------------|----------------|----------|----------------|
| West Plains Fireshed Macrosite  | 1,080          | 98%      | \$1.2M         |
| Batsto Macrosite                | 656            | 84%      | \$4.6M         |
| East Plains Fireshed Macrosite  | 568            | 96%      | \$2.0M         |
| Oswego River Lowlands Macrosite | 119            | 100%     | \$98K          |

Together, these four B1 sites contain **2,423 acres of conflict land valued at \$7.9 million**, with over 90% still vacant. These represent the most ecologically significant and time-sensitive conservation opportunities in the entire state-designated Pinelands.

## Data Center and Large-Scale Development Risk

The Pinelands' large, inexpensive vacant parcels are potentially attractive to data centers and warehouse operators, which seek:

- Large contiguous acreage at low cost
- Proximity to the Atlantic City fiber optic corridor
- Access to regional power infrastructure
- Tax incentive zones in southern New Jersey

The following sites present the highest risk profile for large-scale development:

| Site                                   | Why It Is at Risk                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------|
| <b>West Plains Fireshed Macrosite</b>  | 1,080 acres, 98% vacant at just \$1.2M total, B1 biodiversity rank, large contiguous blocks |
| <b>East Plains Fireshed Macrosite</b>  | 568 acres, 96% vacant, near power corridors, B1 rank                                        |
| <b>Batsto Macrosite</b>                | 656 acres near power/fiber corridors, 84% vacant, B1 rank                                   |
| <b>Forked River Mountain Macrosite</b> | 409 acres, 100% vacant at only \$625K, near existing development in Ocean County            |

Within the state-designated Pinelands, most of these sites fall in Preservation or Forest Areas where the Commission's regulations restrict development. However, parcels can come under pressure if developers seek variances or if management area boundaries are revisited in future Comprehensive Management Plan updates.

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## Regulatory Grey Areas: How Developers Circumvent Pinelands Protections

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The Pinelands Commission's regulatory framework has kept large-scale development out of the region's ecological core for over four decades. But that framework was designed before the data center industry existed. Recent events across southern New Jersey reveal at least seven strategies that developers are already using, or could use, to work around Pinelands protections. Each of these grey areas is grounded in real cases and connects directly to the conflict parcels identified in this analysis.

### 1. The redevelopment designation loophole

Under New Jersey's Local Redevelopment and Housing Law (N.J.S.A. 40A:12A), a municipality can designate an area as "in need of redevelopment" or "rehabilitation" and adopt a redevelopment plan that functions as a zoning overlay. That overlay can permit uses that the underlying zoning would otherwise prohibit, including data centers and large-scale warehouses.

The critical problem is timing. When a town council adopts or amends a redevelopment plan, residents within 200 feet of the affected property are not notified. That level of notice is only required later, when a developer files a specific project application. By that point, the zoning has already been changed, and residents who object are told they are "too late."

This is not hypothetical. In Monroe Township (Gloucester County), the Township Council adopted Ordinance O:02-2025 in December 2025, amending the Hexa Builders Redevelopment Plan to add data centers as a permitted use. When residents raised concerns at the January 2026 planning board

meeting, council members told them the relevant hearings had occurred months earlier and public objections were no longer appropriate. In Vineland, a 2.6-million-square-foot data center was partially built and operating before its environmental impact statement was published in February 2026.

Within the state-designated Pinelands, the Commission must approve changes to municipal redevelopment plans that affect Pinelands lands. This provides a backstop that municipalities outside the Pinelands do not have. But the redevelopment designation itself can still be adopted quietly before residents are aware, and the Commission's review occurs after the municipal decision, not before it. Any of the 331 conflict parcels in this analysis, particularly the 19 in Regional Growth Areas, could be swept into a redevelopment designation without advance public notice.

## **2. The warehouse-to-data center disguise**

Zoning codes in many municipalities do not define “data center” as a distinct land use. A developer can file an application for a “warehouse” or “distribution center,” which faces lighter scrutiny and fits within a broader range of permitted uses, and then operate the building as a data center.

This strategy surfaced in Monroe Township, where Hexa Builder LLCs received preliminary approval in January 2026 for a two-warehouse complex totaling 1.6 million square feet. Residents and local officials suspect the true intended use is a data center, based on the building's design, utility requirements, and the township's recent decision to add data centers to the same redevelopment plan. The Pinelands Commission is currently reviewing the application as a warehouse (Application #1987-0276.012).

For the Pinelands conflict parcels, this means that the 15 existing commercial/industrial parcels (844 acres) are not the only ones at risk. A developer could rezone vacant or agricultural land through a redevelopment designation, file under a warehouse classification, and build a facility that functions as a data center without the project ever being evaluated as one.

## **3. Wetland buffer reduction**

The Pinelands Commission prohibits development within wetlands and within a 300-foot buffer around them. This is one of the Commission's most powerful tools. In our analysis, wetland buffers alone eliminated 275 of 331 conflict parcels (83%) from genuine development risk, including all 19 conflict parcels in the Regional Growth Area.

But the buffer is not absolute. Developers can petition the Commission to reduce it. Heidi Yeh, PhD, the Policy Director for the Pinelands Alliance who assisted in guiding this analysis, noted directly: “Sometimes developers convince the Pinelands Commission to reduce that buffer slightly.” Each reduction is decided case by case, but every approved reduction establishes precedent for the next request.

The stakes are significant. The 275 conflict parcels shielded by wetland buffers represent approximately 4,345 acres. Even a modest, targeted reduction from 300 feet to 200 feet around specific parcels could re-expose dozens of acres of headwater and heritage land to development, particularly in the Regional Growth Area where the Commission otherwise encourages building.

#### **4. Waivers of strict compliance**

The Comprehensive Management Plan includes a formal mechanism called a “Waiver of Strict Compliance” that allows the Commission to grant exceptions to CMP requirements on a case-by-case basis. Waivers are intended for situations where strict application of the rules would create an extraordinary hardship, but the process is discretionary and each approval sets a benchmark for future applications.

Pinelands Development Credits (PDCs) add another dimension. The CMP allows PDCs to be used “in association with municipal variances in Regional Growth Areas, Pinelands Villages and Pinelands Towns, as well as for waivers of strict compliance approved by the Commission in any Pinelands management area.” This means a waiver is not limited to the growth-oriented zones. A developer could, in theory, seek a waiver in a Preservation Area or Forest Area and use PDCs to support the application. The Commission maintains a public tracking system for active waiver applications, but the bar for denial is legally high and the process is not widely followed by affected residents.

#### **5. The public utility exemption**

Under N.J.S.A. 40:55D-19, facilities classified as “public utility” infrastructure can receive an exemption from municipal land use review through the New Jersey Board of Public Utilities (BPU). When a project receives a BPU exemption, the municipality cannot apply its normal zoning requirements.

Data centers are increasingly marketed as “critical digital infrastructure.” While a data center itself may not qualify as a public utility, the power generation and transmission facilities that serve it might. The Vineland DataOne project illustrates how this works in practice: the original data center approval was followed by a separate amendment (Vineland Planning Board Resolution 6720, July 2025) adding three dedicated power generation buildings. If those generation facilities were structured as utility assets, they could potentially claim BPU exemption from local review.

The Pinelands Commission has acknowledged a regulatory gap here. When a utility project receives a BPU exemption from municipal review, the CMP’s coordinated permitting provisions (N.J.A.C. 7:50-4.81 through -4.85) are supposed to ensure the project still receives Commission review. But the gap between the exemption and the Commission’s ability to intervene is real, and a sophisticated developer could exploit the ambiguity around what constitutes “utility infrastructure” to move supporting facilities through the BPU pathway rather than through normal Pinelands review.

#### **6. Phased and incremental development**

Breaking a large project into smaller phases allows a developer to avoid triggering review thresholds that would apply to the full build-out. Each individual phase may appear to comply with management area restrictions, even if the cumulative project would not.

The Vineland DataOne facility demonstrates this approach clearly. The project was approved in four phases over twelve years, totaling 2,647,628 square feet. The power generation buildings were added as a separate amendment months after the original approval. No single phase triggered the level of scrutiny that a 2.6-million-square-foot, 350-megawatt campus would have received if reviewed as one application.

In the Pinelands, phased development is particularly dangerous because the CMP evaluates applications based on the specific proposal submitted, not on the developer's long-term plans. A developer could file for a modest first phase on a conflict parcel in the Forest Area, receive approval, and then submit successive expansions, each building on the precedent of the last. By the time the project reaches its intended scale, the Commission has already approved the foundational use.

## 7. Edge-of-boundary siting

Perhaps the simplest strategy is to build just outside the state-designated Pinelands Area, beyond the Commission's jurisdiction, while still drawing on Pinelands water, generating noise and light that penetrates into the Pinelands, and fragmenting migration corridors that cross the boundary.

Vineland is the clearest example. It sits outside the Pinelands Commission's jurisdiction but directly on top of the Kirkwood-Cohansey aquifer, the same aquifer that sustains every stream, wetland, and cedar swamp within the Pinelands. A 350-megawatt data center drawing millions of gallons of cooling water per day from this aquifer has the potential to lower water tables across the region, with effects that do not stop at the Commission's boundary line.

The noise analysis in Part II of this report reinforces this concern. A data center's acoustic footprint extends over two kilometers beyond the property line. A facility sited one mile outside the Pinelands boundary would produce noise levels above 45 dB inside the Pinelands, sufficient to disrupt barred owl hunting, degrade bat foraging, and mask treefrog breeding calls. The NJDEP is responsible for coordinating with the Pinelands Commission to protect resources outside the Pinelands that affect the Kirkwood-Cohansey aquifer, but no comparable coordination mechanism exists for noise, light, or habitat fragmentation.

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Part II of this report examines a dimension of large-scale industrial development that traditional parcel-level impact assessment does not account for: acoustic pollution. The findings show that data center noise would extend ecological damage far beyond any property line, affecting areas many times larger than the parcels themselves.

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# Part II: The Sound of Development — Data Center Noise and Wildlife

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## The Quiet Pinelands

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The New Jersey Pinelands is one of the quietest landscapes remaining on the Eastern Seaboard. On a still night in the Preservation Area, the only sounds are wind through pitch pines, the nasal "quonk" of a treefrog calling from a cedar swamp, and the distant hoot of a barred owl.

That acoustic environment is not just pleasant. It is essential. Dozens of Pinelands species depend on sound to find mates, locate prey, defend territories, and detect predators. When industrial noise enters a quiet ecosystem, it does not simply add a layer of annoyance. It disrupts the biological processes that sustain populations across a much wider area than the development itself.

This section examines what would happen to Pinelands wildlife if data centers — among the noisiest industrial facilities currently proposed in southern New Jersey — were built on the 331 conflict parcels identified in Part I. The findings demonstrate that noise extends the ecological damage of development far beyond property lines, affecting up to **59% of all heritage biodiversity habitat** in the state-designated Pinelands.

## How Loud Is a Data Center?

Data centers are not typical office buildings. They house thousands of servers that generate enormous heat, requiring industrial-scale cooling systems that run 24 hours a day, 365 days a year. The primary noise sources are:

| Source                      | Noise Level (at source)    | Character                     |
|-----------------------------|----------------------------|-------------------------------|
| Cooling towers and chillers | 65-85 dB(A)                | Continuous low-frequency hum  |
| Backup diesel generators    | 95-105 dB(A)               | Periodic high-intensity roar  |
| Electrical transformers     | 50-65 dB(A)                | Continuous low-frequency buzz |
| Combined boundary noise     | ~70 dB(A) at property line | 24/7, year-round              |

For context, 70 dB(A) is roughly the volume of a vacuum cleaner or busy restaurant. In the interior Pinelands, where nighttime ambient noise drops below 30 dB(A), a data center represents a 10,000-fold increase in sound energy at the property line.

Sound attenuates with distance, but it carries far in the flat, forested Pinelands terrain:

| Distance from facility | Estimated noise level | Ecological threshold                  |
|------------------------|-----------------------|---------------------------------------|
| Property line          | ~70 dB(A)             | All wildlife displaced                |
| 200m                   | ~55 dB(A)             | Amphibian call masking begins         |
| 500m                   | ~50 dB(A)             | Amphibian breeding severely disrupted |
| 1 km (0.6 mi)          | ~45 dB(A)             | Bird and bat communication impaired   |
| 2 km (1.2 mi)          | ~35 dB(A)             | Approaching natural Pinelands ambient |

## The Species at Risk

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The Pinelands is home to 43 animal species listed as threatened or endangered by the NJ Division of Fish and Wildlife. Several of these species are directly dependent on the acoustic environment that data center development would destroy.

### **Pine Barrens Treefrog (*Hyla andersonii*) — State Threatened**

The Pine Barrens treefrog is the iconic amphibian of the Pinelands, a small, bright green frog with a lavender racing stripe that breeds exclusively in acidic, tea-colored wetlands. It is found in only three isolated regions of the eastern United States, with the New Jersey Pinelands population being the largest and most stable.

**Why noise is lethal to this species:** Male Pine Barrens treefrogs attract mates by producing a distinctive nasal call from the edges of breeding pools on humid June evenings. Females select mates based on call quality, favoring louder, more persistent callers. This acoustic mating system is the sole mechanism by which the species reproduces.

Research on related treefrogs (the grey treefrog, *Hyla chrysoscelis*, and the Mundo Novo treefrog, *Boana marginata*) demonstrates that traffic noise at just 55 dB elevates female response thresholds by 20-25 dB, effectively masking male calls and preventing females from locating mates. At 65 dB, call rates decrease, pulse patterns change, and breeding chorus structure collapses.

**At our modeled 500m noise zone (~55 dB), 27,884 acres of heritage habitat would be affected.** Many of the top conflict sites, including the West Plains Fireshed Macrosite, East Plains Fireshed Macrosite, and Batsto Macrosite, are known Pine Barrens treefrog habitat. A single data center in any of these locations could suppress treefrog reproduction across thousands of surrounding acres of otherwise protected breeding habitat.

### **Southern Gray Treefrog (*Hyla chrysoscelis*) — State Endangered**

The southern gray treefrog, a close relative of the Pine Barrens treefrog, is listed as endangered in New Jersey. Like its cousin, it depends entirely on acoustic signaling for reproduction. Research conducted directly on this species confirmed that road noise at 55-65 dB masks advertisement calls and reduces female approach behavior. This species would face the same breeding disruption as the Pine Barrens treefrog within the 500m noise zone.

### **Barred Owl (*Strix varia*) — State Threatened**

The barred owl is the Pinelands' largest resident owl, a species that requires extensive tracts of mature, undisturbed wetland forest. Its territory typically spans 0.4 to 3.4 km<sup>2</sup>, and it relies heavily on vocalizations for mate attraction, territorial defense, and pair bonding.

**Why noise threatens this species:** Barred owls are documented to avoid areas within approximately 500 meters of consistent human activity and noise sources. Their territorial calls operate in the 400-800 Hz frequency range, directly overlapping with the low-frequency hum produced by data center cooling systems. Industrial noise in this frequency band would mask territorial calls, potentially causing territory abandonment, failed pair bonding, and reduced nesting success.

**At our modeled 1km noise zone (~45 dB), 58,153 acres of heritage habitat would be exposed.** This represents **one-third of all heritage biodiversity habitat** in the state-designated Pinelands. Barred owls in these areas would experience chronic interference with their vocal communication throughout the breeding season and year-round territorial maintenance.

### **Northern Long-Eared Bat (*Myotis septentrionalis*) — State and Federally Endangered**

The northern long-eared bat was confirmed in the Pinelands during 2021 surveys. Already devastated by White-Nose Syndrome (a fungal disease that has killed over 6 million bats nationwide since 2006), this species hunts by emitting echolocation pulses at approximately 20 per second while searching and up to 200 per second during prey capture. It forages for moths, beetles, caddisflies, and leafhoppers, consuming more than half its body weight nightly.

**Why noise threatens this species:** Peer-reviewed research published in *Proceedings of the Royal Society B* demonstrates that anthropogenic noise directly “distracts foraging bats,” reducing prey detection and capture rates. Studies on multiple bat species confirm that noise masks echolocation return signals, forcing bats to either increase call intensity (expending energy), shift to less efficient foraging strategies, or abandon noisy areas entirely. Traffic noise has been shown to reduce bat foraging efficiency measurably at distances well beyond the road corridor.

Data center cooling systems produce broadband noise that includes components in the ultrasonic range used by echolocating bats. A facility operating 24/7 would create a permanent dead zone for bat foraging across its noise footprint, precisely in the headwater and wetland corridors where insect prey concentrations are highest.

### **Eastern Tiger Salamander (*Ambystoma tigrinum*) — State Endangered**

The eastern tiger salamander is one of New Jersey's rarest amphibians, breeding in isolated temporary pools in the Pinelands. While salamanders do not rely on acoustic signaling for mating in the same way treefrogs do, they are sensitive to ground vibration from industrial equipment. Data center construction (pile driving, heavy equipment) and operation (generator vibration, compressor cycling) would produce substrate-borne vibrations that could disturb underground larval and adult habitat near breeding pools.

### **Blue-Spotted Salamander (*Ambystoma laterale*) — State Endangered**

Like the tiger salamander, the blue-spotted salamander breeds in fishless, temporary wetland pools within mature forests. The species is extremely sensitive to habitat disturbance. The construction phase of data center development, including road building, grading, and stormwater management, would fragment the forested corridors these salamanders use for migration between breeding pools and upland habitat. The operational noise footprint would add a chronic stressor to already fragile populations.

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## The Numbers: How Far Noise Reaches

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Our spatial analysis buffered all 331 conflict parcels at three noise attenuation distances and measured the overlap with ecologically sensitive areas.

### Heritage biodiversity site impact

| Noise Zone                            | Heritage Acres Affected | % of All Heritage Habitat |
|---------------------------------------|-------------------------|---------------------------|
| Parcel footprint only                 | 3,086                   | 1.8%                      |
| 500m (~55 dB, amphibian call masking) | 27,884                  | 16.0%                     |
| 1km (~45 dB, bird/bat disruption)     | 58,153                  | 33.4%                     |
| 2km (~35 dB, approaching ambient)     | 103,255                 | 59.3%                     |

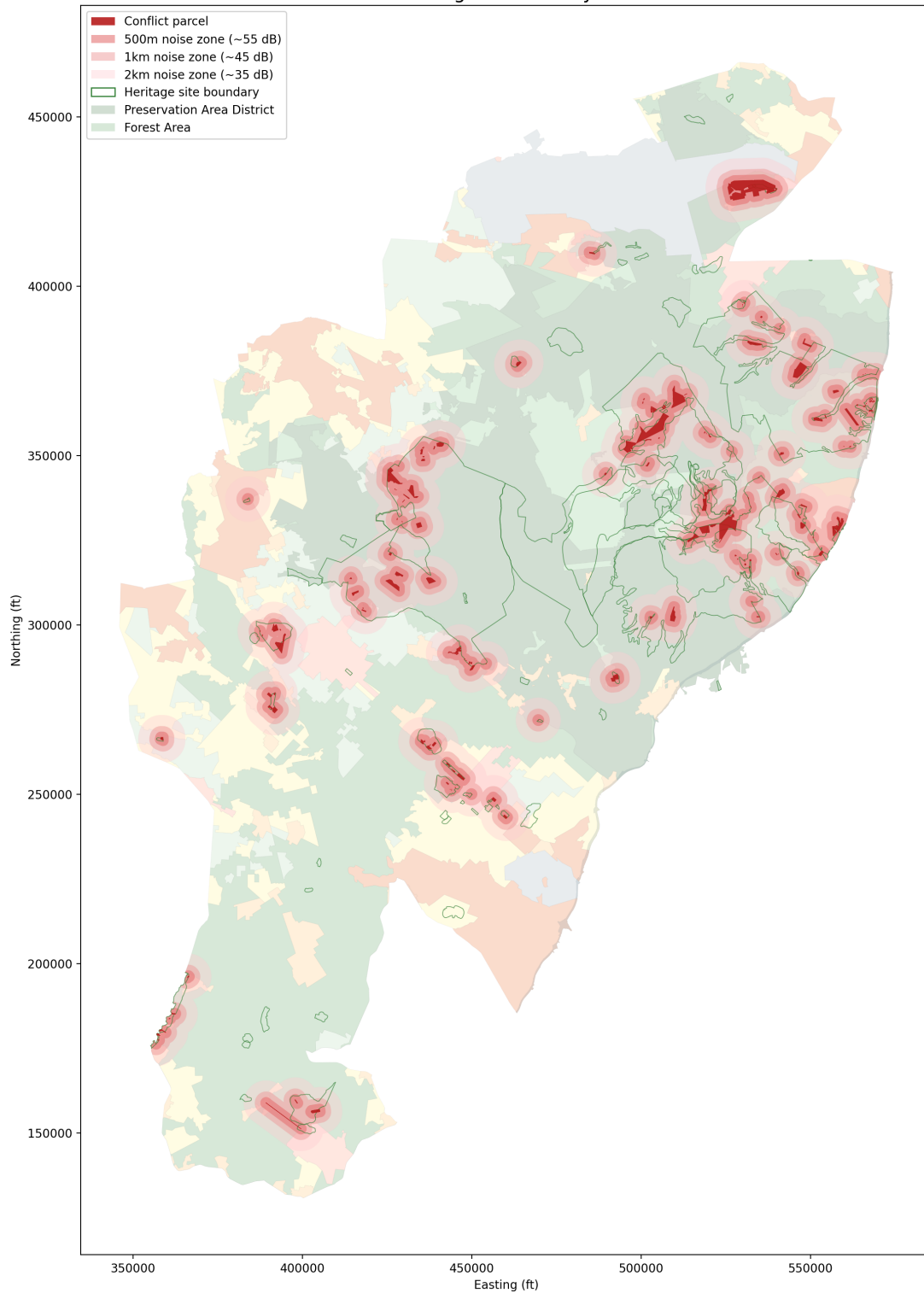
The parcel footprint alone affects less than 2% of heritage habitat. But noise at the 1km threshold — the level at which bird and bat communication is impaired — pushes that impact to **one-third of all heritage habitat** in the state-designated Pinelands. At the 2km threshold, where noise still exceeds natural ambient levels, nearly **60% of all heritage habitat** is affected.

### Headwater zone impact

| Noise Zone            | Headwater Acres Affected | % of All Headwater Area |
|-----------------------|--------------------------|-------------------------|
| Parcel footprint only | 3,253                    | 1.0%                    |
| 500m (~55 dB)         | 21,506                   | 6.9%                    |
| 1km (~45 dB)          | 40,388                   | 12.9%                   |
| 2km (~35 dB)          | 81,633                   | 26.1%                   |

Headwater streams are where amphibians breed. At 1km, nearly 13% of all headwater area in the Pinelands falls within the noise disruption zone, directly threatening treefrog and salamander breeding habitat.

# Pinelands: Data Center Noise Impact Zones on Heritage Biodiversity Sites



*Pinelands: data center noise impact zones overlaid on heritage biodiversity sites*

## The Noise Multiplier: Impact Far Beyond the Property Line

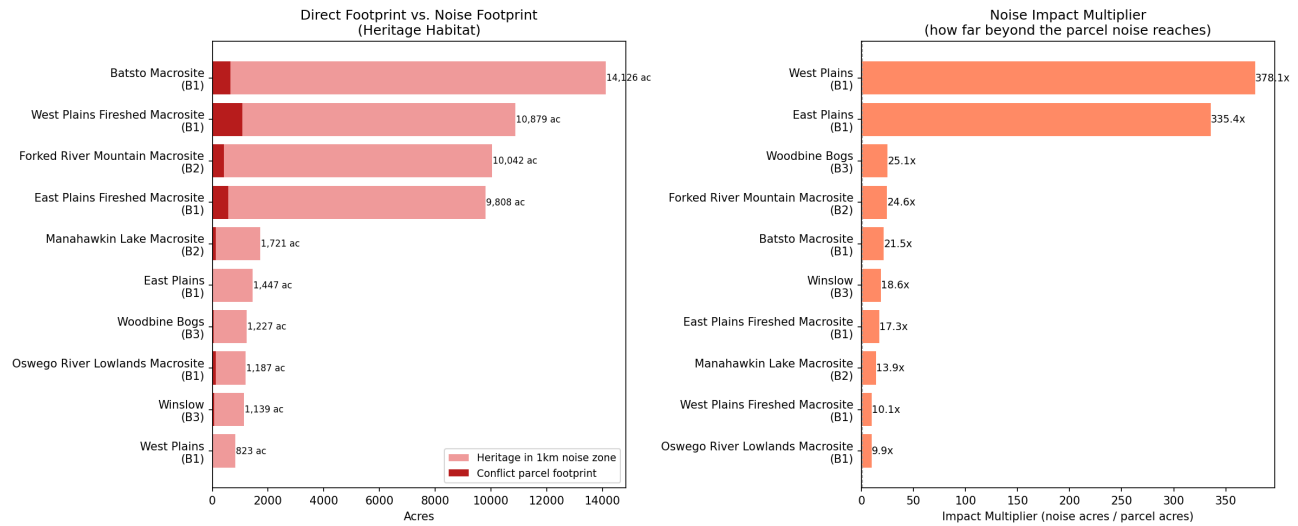
For every acre of conflict parcel, data center noise would degrade many additional acres of surrounding habitat. We calculated this “impact multiplier” for each heritage site:

| Site                                   | Rank      | Conflict Acres | Heritage Acres in 1km Noise Zone | Impact Multiplier |
|----------------------------------------|-----------|----------------|----------------------------------|-------------------|
| <b>Batsto Macrosite</b>                | <b>B1</b> | 656            | 14,126                           | <b>21.5x</b>      |
| <b>West Plains Fireshed Macrosite</b>  | <b>B1</b> | 1,080          | 10,879                           | <b>10.1x</b>      |
| <b>Forked River Mountain Macrosite</b> | B2        | 409            | 10,042                           | <b>24.6x</b>      |
| <b>East Plains Fireshed Macrosite</b>  | <b>B1</b> | 568            | 9,808                            | <b>17.3x</b>      |
| Manahawkin Lake Macrosite              | B2        | 124            | 1,721                            | 13.9x             |
| Woodbine Bogs                          | B3        | 49             | 1,227                            | 25.1x             |
| Oswego River Lowlands Macrosite        | <b>B1</b> | 119            | 1,187                            | 9.9x              |
| Winslow                                | B3        | 61             | 1,139                            | 18.6x             |

At the Batsto Macrosite, a B1-ranked site containing irreplaceable ecological resources, 656 acres of conflict parcels would generate noise affecting **14,126 acres of heritage habitat**, a multiplier of 21.5x. At Forked River Mountain Macrosite, the multiplier reaches 24.6x: just 409 acres of development would noise-pollute over 10,000 acres of protected habitat.

These multipliers demonstrate that traditional parcel-level impact assessment fundamentally understates the ecological cost of data center development in the Pinelands.

## Pinelands: Data Center Noise Extends Far Beyond Parcel Boundaries



Pinelands: direct footprint vs. noise footprint comparison for top heritage sites

## Noise Bleeding Across Management Area Boundaries

When a conflict parcel sits near the edge of its management area, its noise footprint extends into adjacent zones — often areas with much stronger protections. This “noise bleed” represents a regulatory gap: the Pinelands Commission evaluates development applications within a management area, but does not account for noise impacts that cross boundaries.

| Management Area              | Total Acres | Acres in 1km Noise Zone | % Affected | Has Conflict Parcels? |
|------------------------------|-------------|-------------------------|------------|-----------------------|
| Preservation Area District   | 294,338     | 41,630                  | 14.1%      | Yes                   |
| Forest Area                  | 260,119     | 31,560                  | 12.1%      | Yes                   |
| Special AG Production Area   | 37,589      | 6,563                   | 17.5%      | Yes                   |
| Agricultural Production Area | 68,553      | 6,007                   | 8.8%       | Yes                   |
| Regional Growth Area         | 78,399      | 4,415                   | 5.6%       | Yes                   |
| Rural Development Area       | 105,442     | 3,818                   | 3.6%       | Bleed only            |
| Pinelands Village            | 25,937      | 3,814                   | 14.7%      | Yes                   |
| Pinelands Town               | 21,862      | 3,309                   | 15.1%      | Yes                   |

The **Rural Development Area** has no conflict parcels of its own, yet 3,818 acres would be affected by noise bleeding from adjacent areas. This is a protection gap: the Commission cannot regulate noise from a development it never reviewed because it was approved in a neighboring management area.

Most critically, the **Preservation Area District** — the most protected zone in the Pinelands — has 41,630 acres (14.1%) within the 1km noise zone. Noise from development in the Forest Area or Regional Growth Area would penetrate into lands the Commission has specifically designated for maximum ecological protection.

## What the Noise Analysis Reveals

### The traditional impact assessment is incomplete

Current environmental review of development applications in the Pinelands focuses on the parcel itself: what is the land use? Is it in a wetland buffer? Does it have permanent protections? Part I of this report already showed that 331 parcels totaling 5,200 acres pose a direct threat to biodiversity.

The noise analysis reveals that the true impact zone is **10 to 25 times larger** than the parcel footprint. A data center approval on any of these conflict parcels would effectively degrade habitat across thousands of acres of surrounding land, much of it in the most protected management areas.

## Specific consequences by species

| Species                  | Status     | Noise Threshold   | Habitat Affected | Consequence                                  |
|--------------------------|------------|-------------------|------------------|----------------------------------------------|
| Pine Barrens treefrog    | Threatened | 55 dB (500m)      | 27,884 acres     | Breeding call masking, reproductive failure  |
| Southern gray treefrog   | Endangered | 55 dB (500m)      | 27,884 acres     | Breeding call masking, population decline    |
| Barred owl               | Threatened | 45 dB (1km)       | 58,153 acres     | Territory abandonment, nesting failure       |
| Northern long-eared bat  | Endangered | 45 dB (1km)       | 58,153 acres     | Foraging disruption, starvation stress       |
| Eastern tiger salamander | Endangered | Construction zone | Site-specific    | Vibration disturbance, habitat fragmentation |
| Blue-spotted salamander  | Endangered | Construction zone | Site-specific    | Migration corridor disruption                |

## Recommendations

### Immediate priorities

1. **Monitor the 19 Regional Growth Area conflict parcels.** Although wetland buffers and PLP currently cover these parcels, any changes to those protections (such as buffer reductions granted by the Commission) could expose them to development.
2. **Flag B1 sites for proactive monitoring.** The four B1 conflict sites contain 2,423 acres of irreplaceable habitat. Track deed transfers and development applications in these areas.
3. **Coordinate with the Pinelands Commission on the West Plains Fireshed and Batsto Macrosites.** Both are large, high-rank sites with significant vacant conflict acreage that could face development pressure.

## Policy actions

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1. **Advocate for headwater overlay protections.** The Commission's existing wetland buffers are effective (removing 83% of conflict parcels), but headwater streams outside wetland zones receive no comparable protection. A headwater protection overlay would close this gap for the 56 net conflict parcels.
2. **Submit net conflict parcel inventory to the Green Acres program.** The 56 net conflict parcels (855 acres) represent a targeted, affordable acquisition list.
3. **Require noise impact assessments for all large-scale development applications** in the Pinelands, modeled at 500m, 1km, and 2km radii — not just at the property line.
4. **Establish noise overlay zones** around known Pine Barrens treefrog breeding habitat, barred owl nesting territories, and northern long-eared bat roost trees. Development producing continuous noise above 45 dB at these locations should trigger enhanced review.
5. **Close the cross-boundary noise gap.** When a development application in one management area would produce noise affecting a more protective adjacent area, the Commission should evaluate the noise impact under the stricter standard.
6. **Add noise criteria to data center siting guidelines.** Data centers produce uniquely continuous, broadband, 24/7 noise that is qualitatively different from other industrial uses. Siting guidelines should incorporate minimum setbacks from heritage sites and headwater zones based on acoustic modeling.
7. **Request enhanced review for large-scale commercial development in headwater/heritage zones** from the Commission, particularly if management area boundaries are updated.

## Closing regulatory grey areas

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1. **Support legislation to close the redevelopment loophole.** Senate Bill No. 1857 would exclude farmland from redevelopment and rehabilitation designations. Advocate for amendments that also exclude forested and undeveloped land, preventing municipalities from quietly rezoning Pinelands parcels for industrial use.
2. **Urge the Commission to define “data center” as a distinct land use in the CMP.** Without a formal definition, data centers can be filed as warehouses, distribution centers, or office buildings, bypassing size, noise, and utility thresholds that would otherwise apply.
3. **Oppose wetland buffer reductions on conflict parcels.** The 275 conflict parcels currently shielded by wetland buffers should be flagged so that any buffer reduction request affecting them triggers enhanced ecological review, including headwater and heritage impact analysis.
4. **Require cumulative impact review for phased projects.** When successive applications affect the same site or adjacent parcels, the Commission should evaluate the full planned build-out, not just the phase being submitted.

5. **Advocate for cross-boundary coordination on aquifer and acoustic impacts.** Facilities sited outside the Pinelands but on the Kirkwood-Cohansey aquifer or within acoustic range of heritage sites should be subject to coordinated review between NJDEP and the Commission.

## Ongoing monitoring

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1. **Track land transactions in conflict zones.** Set up deed transfer alerts for the 316 vacant conflict parcels to detect early-stage development interest.
  2. **Monitor active waiver and redevelopment applications.** The Commission's public waiver tracking system and municipal redevelopment agendas should be reviewed regularly for activity involving parcels in or adjacent to conflict zones.
  3. **Fund acoustic monitoring at top conflict sites.** Baseline ambient noise measurements at the Batsto, West Plains Fireshed, East Plains Fireshed, and Forked River Mountain Macrosites would establish the scientific record needed to evaluate future development applications.
  4. **Reassess after management area updates.** When the Commission revises its Comprehensive Management Plan, advocate for conflict parcels to be moved into more protective management areas where appropriate.
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# About the Data and Methods

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## Headwater and Habitat Analysis

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This analysis uses publicly available New Jersey datasets:

- **Parcel boundaries:** NJ Statewide Parcels (NJDEP/NJOIT)
- **Tax classification:** NJ MOD-IV Tax Assessment (NJ Division of Taxation)
- **Stream network:** NHDPlus High Resolution (USGS), flow accumulation and stream network rasters
- **Biodiversity sites:** Natural Heritage Priority Sites (NJDEP Natural Heritage Program)
- **Management areas:** Pinelands Management Areas (NJ Pinelands Commission / NJOGIS, Feb 2026)
- **Permanent land protections:** PLP Restrictions (NJ Pinelands Commission, Oct 2024)
- **Wetlands:** Pinelands Wetlands (NJ Pinelands Commission / NJOGIS), planning-level polygons
- **Study boundary:** State-designated Pinelands Area boundary (NJOGIS, ~940,000 acres), excluding the broader Pinelands National Reserve

Headwater streams were defined as network channels with a drainage area of 1 km<sup>2</sup> or less (flow accumulation <= 1,000 upstream cells at 10m resolution), buffered by 150 meters. Wetlands were buffered by 300 feet per Pinelands Commission regulations. Property values reflect assessed (not

market) values from the most recent MOD-IV tax records. Spatial overlaps were identified using parcel centroid containment. “Net conflict” parcels exclude those within wetland buffers or with permanent land protection restrictions.

The analysis boundary follows the state-designated Pinelands Area as defined by the New Jersey Pinelands Protection Act of 1979 (N.J.S.A. 13:18A). It excludes the Pinelands National Reserve, which includes coastal areas east of the Garden State Parkway and around Atlantic City that are administered by NJDEP rather than the Pinelands Commission.

## Noise Analysis

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Noise attenuation was modeled using standard outdoor point-source decay (geometric spreading with atmospheric absorption), assuming a conservative boundary noise level of 70 dB(A) for a typical data center facility. Noise zones were generated by buffering all 331 conflict parcels at 500m, 1km, and 2km and dissolving into unified zones. Overlap was calculated by intersecting noise zones with heritage biodiversity sites and headwater zones.

Species-specific noise thresholds are drawn from peer-reviewed literature:

- Amphibian call masking at 55 dB: Sun & Bhatt, 2007, *Animal Behaviour*; Lopes et al., 2024, *Acoustics Australia*; Libramento et al., 2025, *Herpetological Conservation and Biology*
- Bat foraging disruption: Luo et al., 2021, *Proceedings of the Royal Society B*; Bunkley et al., 2015, *Ethology*; Siemers & Schaub, 2011, *Proceedings of the Royal Society B*
- Barred owl disturbance buffers: NJDEP Division of Fish and Wildlife species profile; Winton et al., 2020, *Urban Ecosystems*

Noise modeling is conservative: it does not account for topographic channeling, temperature inversions (which trap and amplify sound at night, precisely when nocturnal species are most active), or the cumulative effect of multiple facilities.